Alex Fernsby

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Profile

Data engineer with 7 years of experience building batch and streaming pipelines for retail and logistics companies, most recently as the owner of a shipment-events pipeline carrying 40 million events a day.

Core Competencies

- **Logistics data**: shipment-events pipeline, delivery-performance mart and a Looker explore for operations planners
- **Pipeline orchestration**: Apache Airflow deployments at Tidewater Logistics and Marlow Retail Group, from 45 migrated cron jobs to a 40 million events per day stream
- **Programming**: Python and SQL for ingestion, reporting automation and data cleaning; strong PostgreSQL and BigQuery skills
- **Modelling and warehouse**: incremental dbt models, a star schema for 12 analysts, and support for the move from on-premise PostgreSQL to BigQuery
- **Deployments**: GitHub Actions and Terraform for deployments and environments, plus strong Docker skills
- **Data quality**: Great Expectations tests on 30 core tables and a published dbt test package
- **Collaboration**: data contract with five producing teams, analyst training and mentoring, interviews

Professional Experience

Tidewater Logistics — **Copenhagen**
Senior Data Engineer — *March 2022 - present*

- Owns the shipment-events pipeline: 40 million events per day from Kafka into a PostgreSQL and BigQuery warehouse, orchestrated with Apache Airflow
- Cut the nightly rebuild of the delivery-performance mart from 6 hours to 55 minutes with incremental dbt models
- Introduced data-quality tests with Great Expectations on 30 core tables; failed loads dropped 70 percent in the first quarter
- Defined the shipment-events data contract with five producing teams and wrote the on-call runbook for the pipeline
- Moved the team's deployments to GitHub Actions and Terraform, so a new environment takes 20 minutes instead of two days
- Cut the BigQuery bill for the events warehouse by 35 percent by partitioning and clustering the five largest tables
- Built a Looker explore for the operations team, so planners answer delivery-delay questions themselves instead of filing tickets
- Mentors two junior engineers and leads the weekly data-modelling review

Marlow Retail Group — **Utrecht**
Data Engineer — *June 2018 - February 2022*

- Built the first Airflow deployment at the company and migrated 45 cron jobs onto it
- Wrote Python ingestion services for partner price feeds (REST and SFTP), handling about 2 million rows a day
- Added alerting on the 45 Airflow jobs so late partner feeds are noticed within 15 minutes instead of the next morning
- Designed a customer-orders star schema in PostgreSQL used by 12 analysts
- Supported the move of the warehouse from on-premise PostgreSQL to BigQuery, validating 60 tables row by row
- Trained 12 analysts on dbt and reviewed their first 40 models
- Wrote the migration guide for moving reports off the old warehouse, used by four teams

Delft Energy Cooperative **Delft**
Junior Analyst *September 2016 - May 2018*

- Automated monthly consumption reports with Python and SQL, saving the team about 3 days each month
- Cleaned five years of meter readings with Pandas and documented the 14 data-quality rules that came out of it
- Built a small Flask dashboard that 25 cooperative members used to read their own consumption
- Presented the consumption findings to the cooperative's board twice a year

Education

Delft University of Technology **Delft**
MSc Computer Science (data systems) *2016*

Delft University of Technology **Delft**
BSc Computer Science *2014*

Languages

English: C1

Dutch: Native

German: A2

Projects

Open-source dbt package for freight-rate data tests, 120 GitHub stars, github.com/afernsby-example/freight-tests

2024: Talk "Data contracts in practice", Rotterdam Data Meetup

Certifications

2023: Google Cloud Professional Data Engineer

2022: dbt Analytics Engineering Certification

Publications

2024: "Testing freight data with dbt", Data Engineering Weekly guest post

Honors and Awards

2023: Tidewater Logistics Engineering Excellence Award